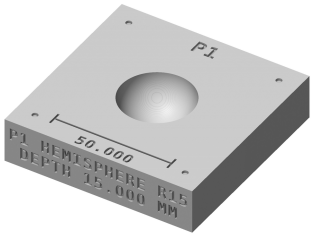
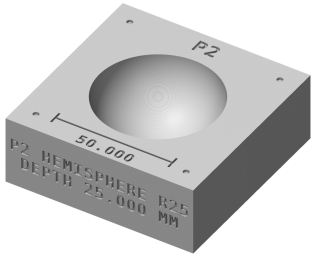
# Wound phantoms: match the part to its picture

Drawn from the STEP files at one scale, cavity up, key line nearest. Sizes in mm, as engraved on the parts.



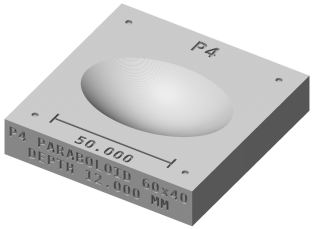

**P1** Hemisphere
Ø30.000
Block 80 × 80 × 19
Depth 15.000 mm
Area 706.86 mm²
Volume 7068.58 mm³
Perimeter 94.25 mm
P1_hemisphere_r15.step



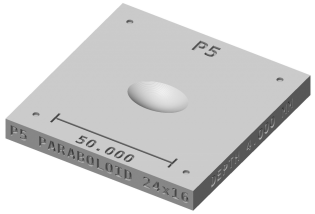

**P2** Hemisphere
Ø50.000
Block 80 × 80 × 29
Depth 25.000 mm
Area 1963.50 mm²
Volume 32724.92 mm³
Perimeter 157.08 mm
P2_hemisphere_r25.step



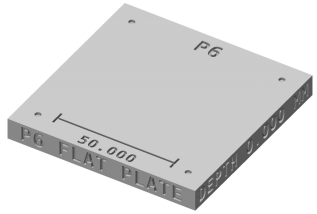

**P3** Cone
Ø40.000
Block 80 × 80 × 14
Depth 10.000 mm
Area 1256.64 mm²
Volume 4188.79 mm³
Perimeter 125.66 mm
P3_cone_r20_d10.step



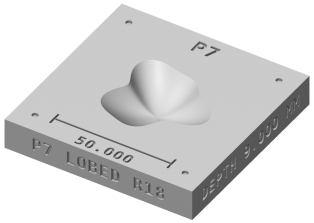

**P4** Paraboloid
60.000 × 40.000 oval
Block 80 × 80 × 16
Depth 12.000 mm
Area 1884.96 mm²
Volume 11309.73 mm³
Perimeter 158.65 mm
P4_paraboloid_30x20_d12.step



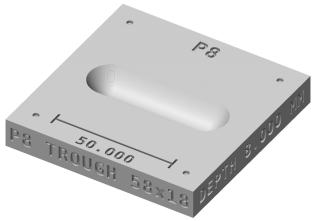

**P5** Paraboloid
24.000 × 16.000 oval
Block 80 × 80 × 9
Depth 4.000 mm
Area 301.59 mm²
Volume 603.19 mm³
Perimeter 63.46 mm
P5_paraboloid_12x8_d4.step



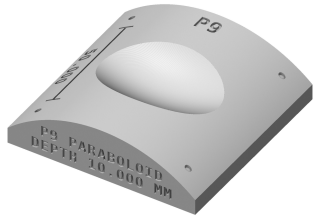

**P6** Flat plate
No cavity
Block 80 × 80 × 9
Depth 0.000 mm
Area 0.00 mm²
Volume 0.00 mm³
P6_flat_plate.step



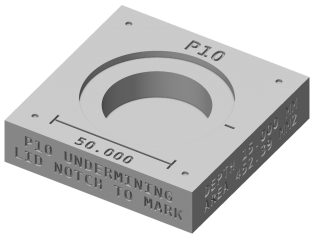

**P7** Lobed
Lobed, R 13.132–22.868
Block 80 × 80 × 13
Depth 9.000 mm
Area 1039.45 mm²
Volume 4677.55 mm³
Perimeter 126.82 mm
P7_lobed_r18_d9.step




**P8** Trough
58.000 × 18.000 slot
Block 80 × 80 × 12
Depth 8.000 mm
Area 974.47 mm²
Volume 4857.88 mm³
Perimeter 136.55 mm
P8_trough_L40_r9_d8.step



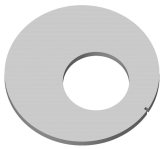

**P9** Paraboloid
50.000 × 36.000 oval
On a curved top, R 60
Block 80 × 80 × 22
Depth 10.000 mm
Plan area 1413.72 mm²
Skin area 1446.60 mm²
Volume 7068.58 mm³
Perimeter 135.98 mm
P9_paraboloid_on_cyl_r60.step




**P10** Undermining
With a lid seated:
Opening Ø24.000
Block 80 × 80 × 19
Depth 15.000 mm
Area 452.39 mm²
Volume 16436.81 mm³
Perimeter 75.40 mm
P10_undermining_base.step



**L1** Lid for P10
Hole Ø24.000
Centred
Disc Ø56 × 3
Undermining:
8.000 all round
Notch to P10’s groove
L1_lid_concentric_r12.step



**L2** Lid for P10
Hole Ø24.000
6.000 off-centre,
on the notch side
Disc Ø56 × 3
Undermining:
2.000–14.000 (page 3)
Notch to P10’s groove
L2_lid_offset6_r12.step

**Revision B prints** have the same shapes and sizes, but look different in four ways: the hollow blocks (all but P5 and P6) have round powder holes through their left and right walls; most of the lettering is missing; P9 has four open slots in its top instead of the dots; and the lids have no notch.

# Every part: file, shape and sizes

Revision C. All dimensions in millimetres. Depth, area and volume are the values engraved on each part, rounded the same way. Revision B prints have the same file names, block sizes and cavities, so they read from this table too; revision B P9 has no fiducial dots (page 3).

| Part | STEP file | How to recognise it | Wound shape (mm) | Opening (mm) | Max depth (mm) | Opening area (mm²) | Perimeter (mm) | Volume (mm³) |
|---|---|---|---|---|---|---|---|---|
| **P1** | P1_hemisphere_r15.step | 19 mm tall; small round bowl, Ø30 | Hemisphere, R 15.000 | Ø30.000 | 15.000 | 706.86 | 94.25 | 7068.58 |
| **P2** | P2_hemisphere_r25.step | 29 mm tall, the tallest block; large round bowl, Ø50 | Hemisphere, R 25.000 | Ø50.000 | 25.000 | 1963.50 | 157.08 | 32724.92 |
| **P3** | P3_cone_r20_d10.step | 14 mm tall; round opening Ø40, straight sides to a point | Cone, R 20.000, h 10.000 | Ø40.000 | 10.000 | 1256.64 | 125.66 | 4188.79 |
| **P4** | P4_paraboloid_30x20_d12.step | 16 mm tall; large oval 60 × 40 | Elliptic paraboloid, a 30.000, b 20.000, h 12.000 | 60.000 × 40.000 oval | 12.000 | 1884.96 | 158.65 | 11309.73 |
| **P5** | P5_paraboloid_12x8_d4.step | 9 mm tall; small, shallow oval 24 × 16 | Elliptic paraboloid, a 12.000, b 8.000, h 4.000 | 24.000 × 16.000 oval | 4.000 | 301.59 | 63.46 | 603.19 |
| **P6** | P6_flat_plate.step | 9 mm tall; flat top, no cavity | None: flat reference | — | 0.000 | 0.00 | — | 0.00 |
| **P7** | P7_lobed_r18_d9.step | 13 mm tall; flower-shaped opening, uneven lobes | Lobed paraboloid, R0 18.000, h 9.000; 3-lobe amplitude 0.18 R0, 5-lobe amplitude 0.10 R0 at phase 0.7 rad | Lobed, R 13.132–22.868 | 9.000 | 1039.45 | 126.82 | 4677.55 |
| **P8** | P8_trough_L40_r9_d8.step | 12 mm tall; long slot 58 × 18, rounded ends | Trough: parabolic channel 40.000 long, R 9.000, h 8.000, half-paraboloid ends | 58.000 × 18.000 slot | 8.000 | 974.47 | 136.55 | 4857.88 |
| **P9** | P9_paraboloid_on_cyl_r60.step | 22 mm tall; top face curved; oval 50 × 36 | Elliptic paraboloid, a 25.000, b 18.000, h 10.000, on a cylinder R 60.000 | 50.000 × 36.000 oval (plan) | 10.000 | 1413.72 skin 1446.60 | 135.98 | 7068.58 |
| **P10** | P10_undermining_base.step | 19 mm tall; stepped round pocket Ø56.4 that the lids drop into | Undermined well: chamber Ø40.000 × 12.000 under a Ø24.000 lid opening | Ø24.000 (the lid's hole) | 15.000 with a lid | 452.39 | 75.40 | 16436.81 |
| **L1** | L1_lid_concentric_r12.step | Disc Ø56 × 3, hole centred | Lid for P10: hole Ø24.000, centred | Sets P10's opening and undermining; values on page 3 | | | | |
| **L2** | L2_lid_offset6_r12.step | Disc Ø56 × 3, hole 6 mm off-centre | Lid for P10: hole Ø24.000, offset 6.000 | Sets P10's opening and undermining; values on page 3 | | | | |

**Blocks** are 80 × 80 mm; the height is in “How to recognise it”. **Max depth** is from the top surface to the lowest point of the wound (on P9, from the crown of its curved top). **Opening area** and **perimeter** are measured in plan, looking straight down; on P9, whose top is curved, the area measured on the curved surface itself is given as “skin”. **Perimeter** is not engraved on the parts. P10’s depth, area, perimeter and volume are with either lid seated (page 3).

Measure the printed parts with calipers on arrival and use the measured values as truth: every print process shrinks a little, and the measured number is the one that matters.

# P10 undermining set

P10 is a Ø40.000 × 12.000 chamber under a Ø56.400 × 3.000 pocket. A lid drops into the pocket, flush with the top, and its Ø24.000 hole becomes the visible wound; the chamber beyond the hole's edge is undermined. The two lids give the same depth, area and volume and differ only in where the undermining runs.

| | P10 with L1 seated | P10 with L2 seated |
|---|---|---|
| **Lid file** | L1_lid_concentric_r12.step | L2_lid_offset6_r12.step |
| **Lid hole** | Ø24.000, centred | Ø24.000, centre 6.000 toward 3 o'clock |
| **Max depth (mm)** | 15.000 (top skin to chamber floor) | 15.000 (top skin to chamber floor) |
| **Opening area (mm²)** | 452.39 | 452.39 |
| **Opening perimeter (mm)** | 75.40 | 75.40 |
| **Total volume (mm³)** | 16436.81 | 16436.81 |
| **Visible from above (mm³)** | 6785.84 | 6785.84 |
| **Hidden under the lid (mm³)** | 9650.97 | 9650.97 |
| **Undermined area (mm²)** | 804.25 | 804.25 |
| **Undermining extent (mm)** | 8.000 all round | 2.000 at 3 o'clock to 14.000 at 9 o'clock |

Without a lid, the chamber alone, measured from the pocket floor 3.000 mm below the top: opening Ø40.000, area 1256.64 mm², depth 12.000 mm, volume 15079.64 mm³.

## Undermining extent at each clock position

From the edge of the lid's hole outward to the chamber wall, along a line from the hole's centre, as a probe would measure it.

| Clock | 12 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **L1 (mm)** | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 | 8.000 |
| **L2 (mm)** | 7.079 | 4.313 | 2.578 | 2.000 | 2.578 | 4.313 | 7.079 | 10.313 | 12.970 | 14.000 | 12.970 | 10.313 |

**Orientation.** 12 o'clock is the edge of P10 nearest the "P10" engraved on its top; 6 o'clock is the side with the 50.000 key line. On revision C parts, seat the lid with the notch in its rim at the index groove on P10's top, which is at 3 o'clock. Revision B parts have no groove, no notch and no legible "P10": take 6 o'clock as the side with the key line, and seat L2 with its hole nearest 3 o'clock. L1 reads the same at every position.

## Scale references on every block's top

**Key line:** 50.000 mm between the centres of its two end ticks. **Fiducial dots:** four Ø2.500 dots at the corners of a 64.000 mm square, centre to centre. Revision B P9 has no dots: its powder vents cut them away and left four open slots in its top near the corners, so use it for depth, area and volume only, not as a scale reference.

## Where the numbers come from

Hemisphere: $V = \frac{2}{3}\pi R^3$, $A = \pi R^2$. Cone: $V = \frac{1}{3}\pi R^2 h$, $A = \pi R^2$. Elliptic paraboloid: $V = \frac{1}{2}\pi abh$, $A = \pi ab$. Lobed paraboloid, $r(\theta) = R0(1 + \Sigma\, a\cos(m\theta + \varphi))$, one term per set of m lobes with amplitude a and phase φ: $V = \frac{1}{4}\pi hR0^2(2 + \Sigma a^2)$, $A = \frac{1}{2}\pi R0^2(2 + \Sigma a^2)$. Trough, channel length L: $V = (4/3)hRL + \frac{1}{2}\pi hR^2$, $A = 2RL + \pi R^2$. P10 with a lid of hole radius r and thickness t: $V = \pi r^2 t + \pi R^2 h$, depth = t + h. Perimeters are integrated numerically from the same outlines.